

Quiz 6: Image Classification of Cats and Dogs using Convolutional Neural Network (CNN)

Problem Description

A dataset containing images of cats and dogs is provided. The dataset consists of 10,000 images, including:

5,000 images of cats

5,000 images of dogs

Using this dataset, perform binary image classification using a Convolutional Neural Network (CNN).

Question 1

CNN Model using 80–10–10 Data Split

- Using the full dataset of 10,000 images, perform a data split of 80%, 10%, and 10% for training, validation, and testing, respectively.
- Develop a CNN model to classify images into cat and dog classes.
- Train the model using the training dataset and monitor model performance using the validation dataset. During the training process, record the training loss and validation loss for each epoch.
- After completing the training process, evaluate the trained model using the test dataset and compute the confusion matrix as well as the F1 score.

Report the following:

- A CSV file containing epoch-wise training loss and validation loss. Upload a CSV file via the google form (quiz). The filename for the csv should be **applicationnumber_largesplit.csv** (example: 123456789_largesplit.csv)
- The confusion matrix obtained from the test dataset. Upload the image file of confusion matrix with the file name as your application number. (Example: Your application number being 120261444; your image file must be seen as 120261444_largesplit.jpg)
- A CSV file containing the confusion matrix table, the column names must be exactly 'Predicted_Cat' and 'Predicted_Dog', and the row names must be exactly 'Actual_Cat' and 'Actual_Dog'. Please make sure you use the exact same row and column labels as specified in the same order. The answers will be checked using automatic code, and if different labels are used, your answers will not be marked. The **F1 score** shall be placed in the third column, with the column name being "F1 Score", then, input the value just below it (in the same third column and second row). Upload a CSV file via the google form (here, quiz 6). The filename for the csv should be applicationnumber_confusion_largesplit.csv (example: 123456789_confusion_largesplit.csv)

Question 2

CNN Model using 50–25–25 Data Split

- Using the full dataset of 10,000 images, perform a data split of 50%, 25%, and 25% for training, validation, and testing, respectively.
- Develop a CNN model to classify images into cat and dog classes.
- Train the model using the training dataset and monitor model performance using the validation dataset. During the training process, record the training loss and validation loss for each epoch.
- After completing the training process, evaluate the trained model using the test dataset and compute the confusion matrix as well as the F1 score.

Report the following:

- A CSV file containing epoch-wise training loss and validation loss. Upload a CSV file via the google form (quiz). The filename for the csv should be **applicationnumber_smallsplit.csv** (example: 123456789_smallsplit.csv)
- The confusion matrix obtained from the test dataset. Upload the image file of confusion matrix with the file name as your application number. (Example: Your application number being 120261444; your image file must be seen as 120261444_smallsplit.jpg)
- A CSV file containing the confusion matrix table as well as F1 score (follow instructions as given in the above question 1). Upload a CSV file via the google form (here, quiz 6). The filename for the csv should be applicationnumber_confusion_smallsplit.csv (example: 123456789_confusion_smallsplit.csv)

Question 3

CNN Model using Reduced Dataset

The original dataset contains 10,000 images (5,000 cats and 5,000 dogs). Reduce the dataset size by half ($x/2$) so that the new dataset contains:

- 2,500 images of cats
- 2,500 images of dogs

The total dataset size will therefore be 5,000 images.

Using this reduced dataset, develop a CNN model for cat and dog classification. Perform an appropriate training, validation, and testing split, train the CNN model, and record the training and validation loss across epochs.

Evaluate the model using the test dataset and compute the confusion matrix as well as the F1 score.

Report the following:

- A CSV file containing epoch-wise training loss and validation loss. Upload a CSV file via the google form (quiz). The filename for the csv should be **applicationnumber_reducedmodel.csv** (example: 123456789_reducedmodel.csv)
 - The confusion matrix obtained from the test dataset. Upload the image file of confusion matrix with the file name as your application number. (Example: Your application number being 120261444; your image file must be seen as 120261444_reducedmodel.jpg)
 - A CSV file containing the confusion matrix table as well as F1 score (follow instructions as given in the above question 1). Upload a CSV file via the google form (here, quiz 6). The filename for the csv should be applicationnumber_confusion_reducedmodel.csv (example: 123456789_confusion_reducedmodel.csv)
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Question 4

Evaluation on a New Dataset

From the models developed in Questions 1–3, identify the best-performing model based on its classification performance.

A new dataset containing 20 images will be provided, consisting of:

- 10 images of cats
- 10 images of dogs

Use the best-performing CNN model to classify these images.

Report the following:

- The confusion matrix obtained from the test dataset as well as the F1 score. Upload the image file of confusion matrix with the file name as your application number. (Example: Your application number being 120261444; your image file must be seen as 120261444_bestmodel.jpg)
 - A CSV file containing the confusion matrix table as well as F1 score (follow instructions as given in the above question 1). Upload a CSV file via the google form (here, quiz 6). The filename for the csv should be applicationnumber_confusion_bestmodel.csv (example: 123456789_confusion_bestmodel.csv)
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Question 5

Evaluation on a New Dataset (same dataset as for question number 4)

From the models developed in Questions 1–3, identify the worst-performing model based on its classification performance.

A new dataset containing 20 images will be provided, consisting of:

- 10 images of cats
- 10 images of dogs

Use the worst-performing CNN model to classify these images.

Report the following:

- The confusion matrix obtained from the test dataset as well as the F1 score. Upload the image file of confusion matrix with the file name as your application number. (Example: Your application number being 120261444; your image file must be seen as 120261444_worstmodel.jpg)
- A CSV file containing the confusion matrix table as well as F1 score (follow instructions as given in the above question 1). Upload a CSV file via the google form (here, quiz 6). The filename for the csv should be applicationnumber_confusion_worstmodel.csv (example: 123456789_confusion_worstmodel.csv).